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(54) **DIFFERENT FORMS OF
6-CHLORO-2-ETHYL-N-(4-(4-(4-(TRIFLUOROMETHOXY)PHENYL)PIPERIDINE-1-YL)BENZYL)IMIDAZO[1,2-A]PYRIDINE-3-CARBOXAMIDE**

VERSCHIEDENE FORMEN VON
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WO-A1-2011/113606 WO-A1-2019/068910

• **PETHE K ET AL: "Discovery of Q203, a potent clinical candidate for the treatment of tuberculosis", NATURE MEDICINE, NATURE PUB. CO, NEW YORK, vol. 19, no. 9, 1 September 2013 (2013-09-01), pages 1157-1162, XP002730638, ISSN: 1078-8956, DOI: 10.1038/NM.3262 [retrieved on 2013-08-04] cited in the application**

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Description

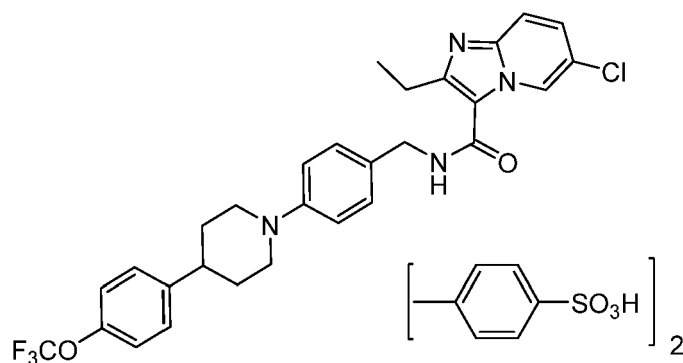
[0001] The present invention relates to different forms of the compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide and to methods of making such forms/com-
pounds. The present invention furthermore relates to mono-acid addition salts thereof, to methods of making such mono-
acid addition salts and to pharmaceutical compositions comprising any of the aforementioned compounds. Furthermore,
the present invention relates to uses of any of these compounds.

[0002] Tuberculosis as a disease continues to result in millions of deaths each year. Inadequate use of chemotherapy has led to an increasing number of drug resistant cases. This situation is likely to worsen with the emergence of extremely resistant strains to all currently known drugs. Current chemotherapy consists of compounds that directly target *Mycobacterium tuberculosis*, either by neutralizing general information pathways and critical processes such as RNA polymerization and protein synthesis inhibition or by interfering with mycobacterial specific cell envelop synthesis. The most widely used dedicated anti-tubercular drugs isoniazid, ethionamide, and pyrazin amide are pro-drugs that first require activation. They are administered to a patient for a course of several months. Patients infected with multi-drug resistant strains of *M. tuberculosis* may have to undergo combination therapies for extended periods of time.

[0003] WO 2011/113606 describes various anti-tubercular compounds and their use in the treatment of bacterial infections, including compound "Q203" which chemically is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide. In a publication by Pethe et al. (Nature Medicine, 19, 1157-1160 (2013)), this compound is reported to be active against tuberculosis by interfering with the bacterial energy metabolism, inhibiting cytochrome bc1 activity which is an essential component of the electron transport chain required for synthesis of ATP.

[0004] Whilst the compound shows promise for future therapy of tuberculosis and related infections, there continues to be a need for forms thereof that are particularly suitable for pharmaceutical administration. In particular there is a need to provide forms that are showing an improved solubility in comparison to the free base of this compound. Furthermore, there is a need in the art to provide for forms that show an improved stability.

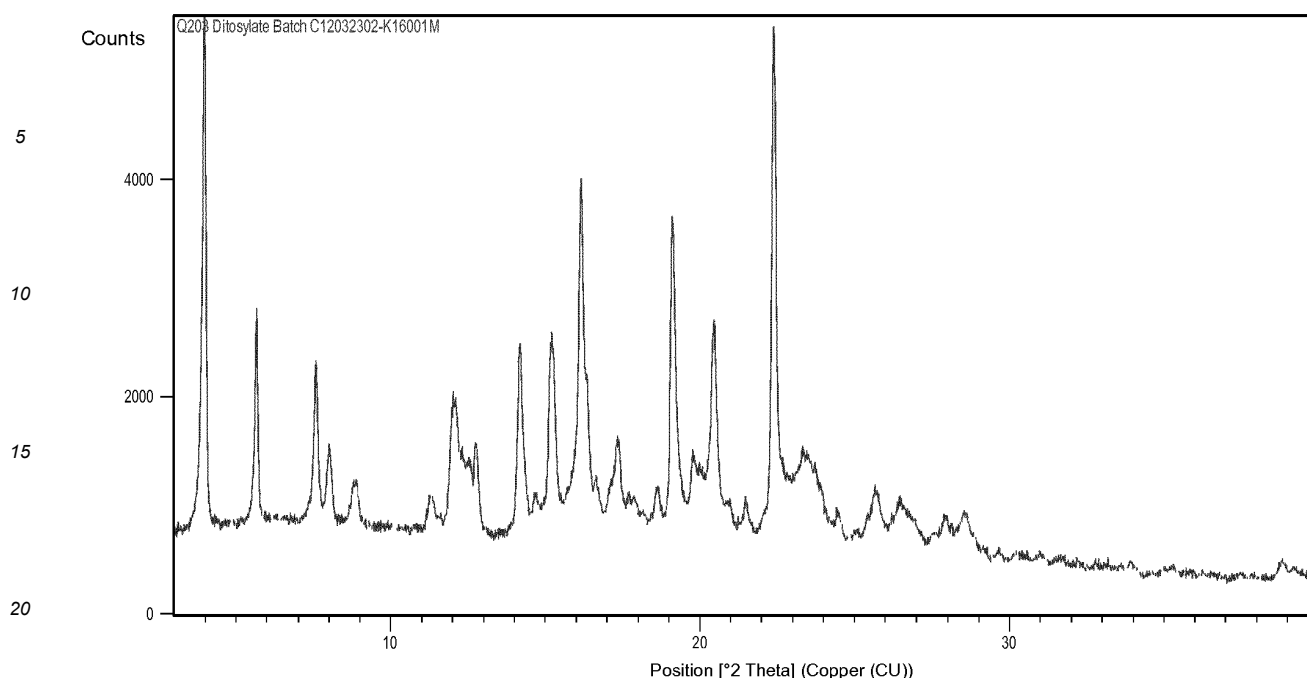
[0005] In a first aspect the present invention relates to a compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate having the structure



and furthermore having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K α - radiation (Cu-K α):

3.9° 2 θ , 5.6° 2 θ , 8.0° 2 θ , 16.1° 2 θ , 19.1° 2 θ , and 22.4° 2 θ , $\pm$ 0.2° 2 θ .

[0006] In one embodiment, the compound has an XRPD spectrum as shown hereafter:



[0007] In one embodiment, the compound has a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of 235°C-237°C.

[0008] In one embodiment, the compound is produced by a method comprising the steps:

- Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2;
- Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone;
- Evaporating the solvent or solvent mixture.

[0009] In a further aspect, the present invention relates to a method for making the compound as defined above, said method comprising the steps:

- Providing, in any order, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2
- Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone;
- Evaporating the solvent or solvent mixture.

[0010] In a further aspect, the present invention relates to a mono-acid addition salt of 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide which is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride, 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate, or 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-tosylate;

which mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-hydrochloride having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K_α - radiation (Cu-K_α):

6.4° 2θ, 8.1° 2θ, 16.2° 2θ, 17.2° 2θ, 24.3° 2θ and 25.0° 2θ, ± 0.2° 2θ; or

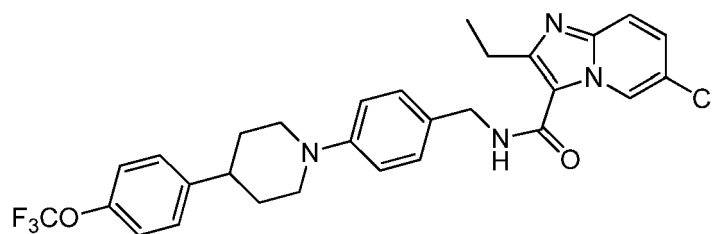
which mono-acid addition salt is 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide mono-phosphate having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K_α - radiation (Cu-K_α):

(continued)

solvents	RT	
	Wet(overnight)	Dry(overnight)
IPA+5%water	A	A
MEK	A	A
MA	A	A
Acetone	A	A
Acetone+5%water	A	A

Example 2

[0072] Inventors wanted to perform a salt/co-crystal screen of Q203 free base, shown below. To this end, a standard salt/co-crystal screen of Q203 was carried out in order to identify salts with acceptable properties and also physicochemical characterisation of Q203 free base was conducted. The starting material used in this example 2 were Q203 free base, batch C12032302-J16001 and a ditosylate salt of Q203, batch C12032302-K16001M.



Chemical Formula: $C_{29}H_{28}ClF_3N_4O_2$

Molecular Weight: 557.01

Elemental Analysis: C, 62.53; H, 5.07; Cl, 6.36; F, 10.23; N, 10.06; O, 5.74

Q203 free base chemical structure

[0073] The following co-formers, listed in Table 1, have been used for the salt/co-crystal of Q203 free base:

Table 1: List of co-formers used in this example 2

Co-formers				
2-furoic acid	isethionic (2-hydroxyethanesulfonic) acid	pamoic acid	hydriodic acid	oxalic acid
ascorbic acid	ketoglutaric (oxoglutaric) acid	phosphoric acid	hydrochloric acid	nitric acid
citric acid	lactobionic acid	p-toluenesulfoni c acid	glycolic (hydroxyacetic) acid	naphtalene-2-sulfonic acid
estolic (laurylsulfonic) acid	maleic acid	pyroglutamic (L) acid	glucoheptonic acid	mono-methyl ester of sulfonic acid
ethane-1,2-disulfonic acid	malic (L) acid	pyruvic (2-oxopropanoic) acid	glucaric (saccharic) acid	methanesulfoni c acid

Table 21: Experimental conditions leading to Type 41 solid (pure and mixtures)

Salt former	Sample No. (ND-0006E-)	Acid:API (molar ratio)	Result	XRPD
lactobionic	010-01	1:1 molar eq.	solid	Type 41
ascorbic	010-07	1:1 molar eq.	solid	Type 41
lactobionic	008-01	1:1 molar eq.	solid	Type 41
2-furoic	010-14	1:1 molar eq.	solid	Type 41 + T28
galactaric (mucic)	010-01	1:1 molar eq.	solid	Type 41 + T28

5.42 Type 42 (*gentisic*)

[0165] Type 42 material was isolated from a LAG experiment using gentisic and Q203 free base (1:1 acid/API) in acetone. XRPD analysis of Type 42 solid (ND-0006E-008-08, Figure 65) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed no peak shifting, ~0.14 molar eq. of acetone (~1.12% w/w) and ~0.67 molar eq. of co-former (data not shown). TG/DTA showed a weight loss from 25 to 113°C (likely moisture) and an endotherm was observed at onset around 121°C. These results suggest either a polymorph or a potential co-crystal of Q203 free base. Further analysis would be needed to confirm one of these assumptions.

5.43 Type 43 (*ascorbic*)

[0166] Type 43 material was isolated from a LAG experiment using ascorbic and Q203 free base (1:1 acid/API) in a mixture of IPA/water. XRPD analysis of Type 43 solid (ND-0006E-008-02, Figure 66) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting as expected, no residual IPA and ~0.8 molar eq. of co-former. TG/DTA showed a weight loss from 25 to ~131°C of ~0.9% likely attributable to water moisture. Two endotherm events were observed at onset of 136 and 158°C associated with a constant weight loss of ~11% from 130 to 300°C. These results suggest that Type 43 may be a mixture of polymorphs (free base and co-former) or a potential co-crystal of Q203. Further analysis would be required to better understand the nature of this Type.

5.44 Type 44 (*saccharin*)

[0167] Type 44 material was isolated from a LAG experiment using saccharin and Q203 free base (1:1 acid/API) in a mixture of IPA/water. XRPD analysis of Type 44 solid (ND-0006E-008-14, Figure 67) showed the material was disordered crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed peak shifting with ~0.5 molar eq. of IPA suggesting IPA hemisolvate of Q203 saccharin salt. Further experiments may lead to unsolvated saccharin salt of Q203

5.45 Type 45 (*oxalic*)

[0168] Type 45 material was isolated from a HT slurry experiment (40C) using oxalic and Q203 free base (1:1 acid/API) in acetone for 5 days. XRPD analysis of Type 45 solid (ND-0006E-010-04, Figure 68) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis showed peak shifting with ~0.06 molar eq. of acetone suggesting salt formation of Q203 oxalate salt (data not shown).

5.46 Type 46 (*oxalic*)

[0169] Type 46 material was isolated from a HT slurry experiment (40C) using oxalic and Q203 free base (1:1 acid/API) in methanol for 5 days. XRPD analysis of Type 46 solid (ND-0006E-010-05, Figure 69) showed the material was crystalline. A slight baseline drift suggests the presence of amorphous content. Proton NMR analysis (data not shown) showed no peak shifting suggesting polymorph of the free base.

(continued)

Time [months]	Purity [%] (as measured by HPLC)
36	>98%
48	>98%
60	>98%

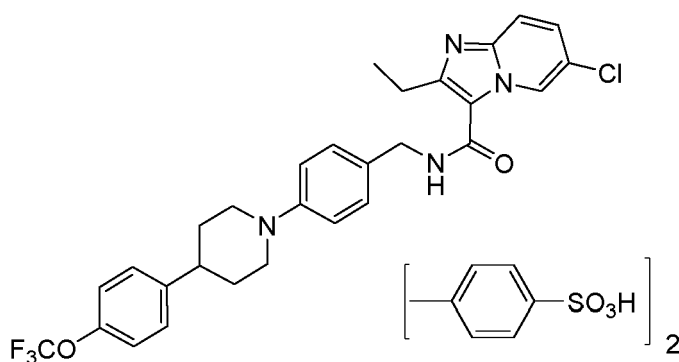
[0199] It can be seen that the purity remains >98% and does not show any sign of decomposition or deterioration.

[0200] Furthermore, in the experiments under accelerated conditions, there was no decomposition for a period up until 6 months when the sample still showed a purity of >98% (data not shown).

[0201] From these data it can be concluded that Q203-ditosylate form A is stable under long-term storage conditions of 25°C and 60% RH for up to 60 months and under accelerated conditions of 40°C and 75% RH up to 6 months.

Claims

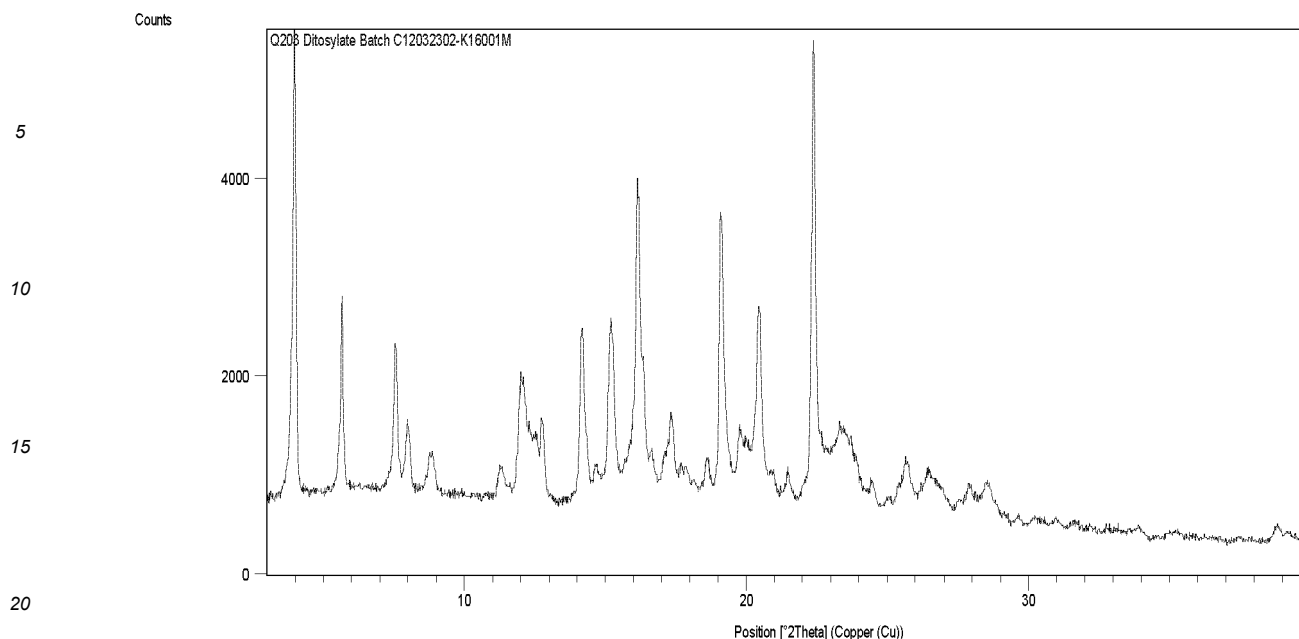
1. Compound 6-chloro-2-ethyl-N-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-a]pyridine-3-carboxamide ditosylate having the structure



and furthermore having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu-K α - radiation (Cu-K α):

3.9° 2 θ , 5.6° 2 θ , 8.0° 2 θ , 16.1° 2 θ , 19.1° 2 θ , and 22.4° 2 θ , $\pm$ 0.2° 2 θ .

2. The compound according to claim 1 having an XRPD spectrum as shown hereafter:



3. The compound according to any of the foregoing claims having a differential scanning calorimetry (DSC) thermogram showing a single endotherm peak with an onset of 235°C-237°C.

4. The compound according to any of the foregoing claims, being produced by a method comprising the steps:

- Providing, in any order, 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2;
- Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone;
- Evaporating the solvent or solvent mixture,

5. A method for making the compound as defined in any of claims 1-4, said method comprising the steps:

- Providing, in any order, 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide free base and para-toluenesulphonic acid in a stoichiometric ratio of 1:2;
- Mixing and dissolving them in a suitable solvent or solvent mixture, such as isopropylalcohol (IPA), tetrahydrofuran (THF), acetone or a mixture of THF and acetone;
- Evaporating the solvent or solvent mixture.

6. A mono-acid addition salt of 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide which is 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide mono-hydrochloride, 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide mono-phosphate, or 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide mono-tosylate;

which mono-acid addition salt is 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide mono-hydrochloride having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu- K_{α} - radiation (Cu- K_{α}):

6.4° 2 θ , 8.1° 2 θ , 16.2° 2 θ , 17.2° 2 θ , 24.3° 2 θ and 25.0° 2 θ , $\pm$ 0.2° 2 θ ;

or

which mono-acid addition salt is 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)imidazo[1,2-*a*]pyridine-3-carboxamide mono-phosphate having the following peaks in an X-ray Powder Diffraction (XRPD) spectrum obtained by irradiation with Cu- K_{α} - radiation (Cu- K_{α}):

9.0° 2 θ , 10.7 $\pm$ 0.2° 2 θ , 11.7° 2 θ , 14.8° 2 θ , 18.4° 2 θ , 19.3° 2 θ , and 21.8° 2 θ , 22.8° 2 θ , $\pm$ 0.2° 2 θ ;

or

which mono-acid addition salt is 6-chloro-2-ethyl-*N*-(4-(4-(4-(trifluoromethoxy)phenyl)piperidine-1-yl)benzyl)im-